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College of Computing and Information Science

Deep Learning

Week 7: Diffusion Models

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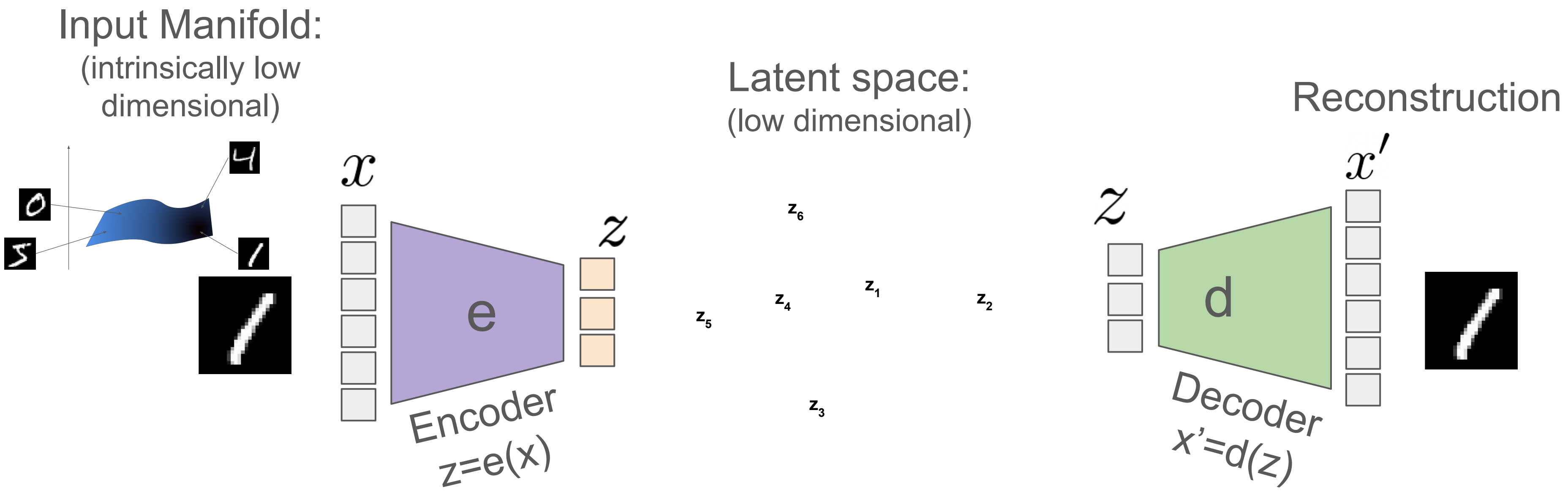
VAE Recap

Auto-Encoder

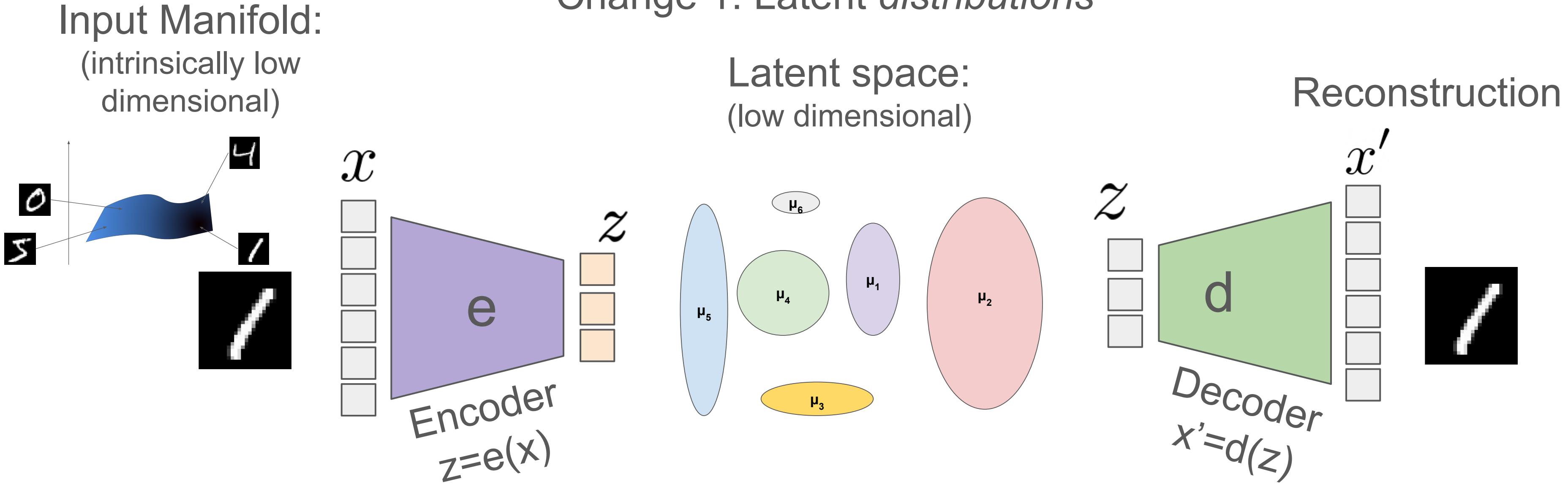


Reconstruction Loss:

$$\min_{\phi,\theta} \sum_{\mathbf{x}\in\mathcal{D}} (\mathbf{x} - d_\theta(e_\phi(\mathbf{x})))^2$$

Auto-Encoder

Change 1: Latent *distributions*



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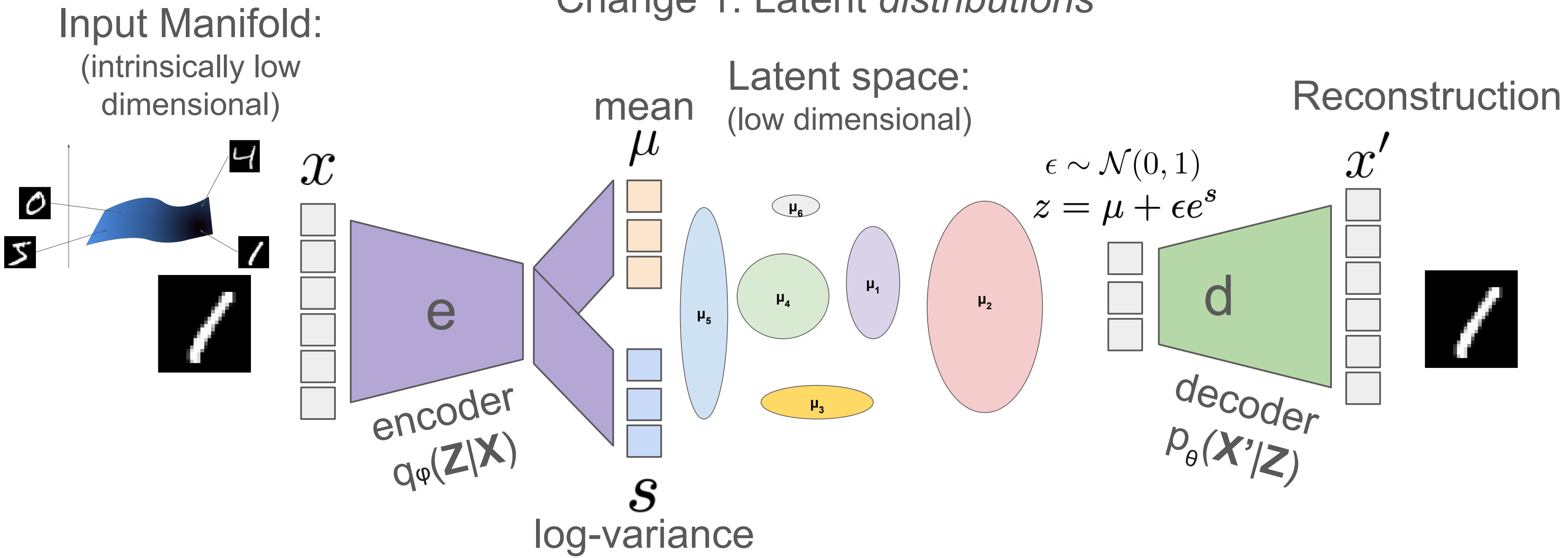
$$\min_{\phi,\theta} \sum_{\mathbf{x} \in \mathcal{D}} (\mathbf{x} - d_\theta(e_\phi(\mathbf{x})))^2$$

Auto-Encoder

Change 1: Latent *distributions*

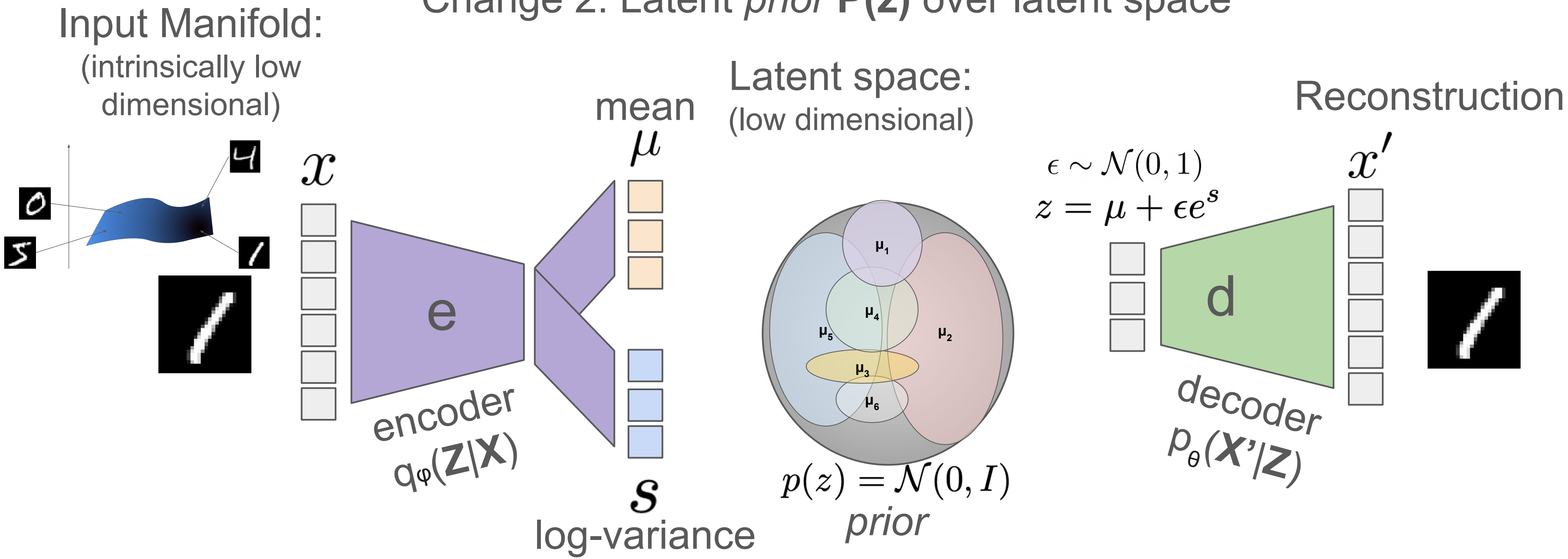


Reconstruction Loss:

$$\min_{\phi,\theta} \mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[(\mathbf{x} - d_\theta(\mathbf{z}))^2\right]$$

Auto-Encoder

Change 2: Latent *prior* **P(z)** over latent space



$$\min_{\phi,\theta} \underbrace{\mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[(\mathbf{x} - d_\theta(\mathbf{z}))^2\right]}_{\text{reconstruction term}} + \underbrace{D_{\mathrm{KL}}\left(q_\phi(\boldsymbol{z}|\boldsymbol{x}) \| p(\boldsymbol{z})\right)}_{\text{prior matching term}}$$

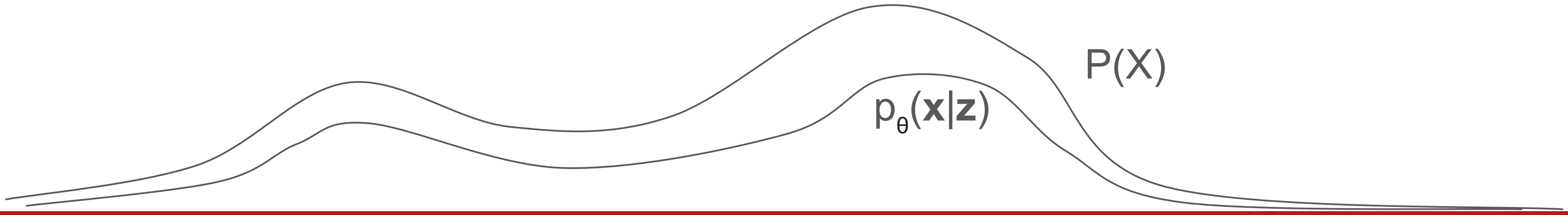
Evidence Lower Bound (ELBO)

First note that we can minimize reconstruction loss or maximize rec. likelihood:

$$\min_{\phi,\theta} \underbrace{\mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[(\mathbf{x}-d_\theta(\mathbf{z}))^2\right]}_{\text{reconstruction term}} + \underbrace{D_{\mathrm{KL}}\left(q_\phi(\boldsymbol{z}|\boldsymbol{x})\|p(\boldsymbol{z})\right)}_{\text{prior matching term}} = \max_{\phi,\theta} \underbrace{\mathbb{E}_{z\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}\left(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})\right)}_{\text{prior matching term}}$$

The ELBO proves that we are maximizing a lower bound of the data distribution:

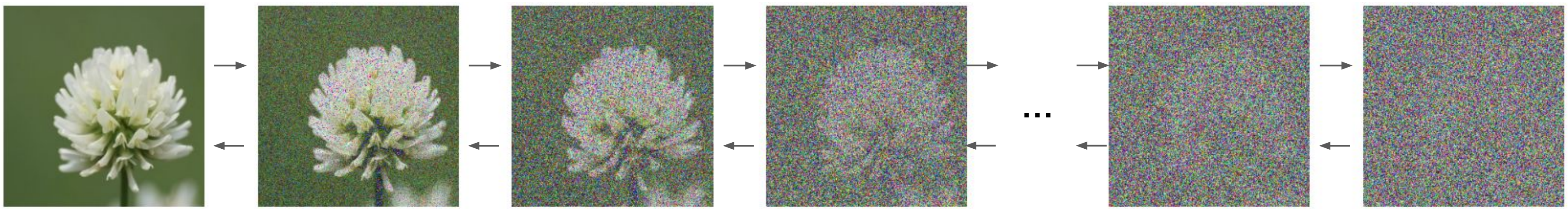
$$\log(P(x)) \geq \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}\left(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})\right)}_{\text{prior matching term}}$$



Diffusion Models

Denoising Diffusion Models

Forward Process (think encoder) [fixed]
Destroy by successively adding Gaussian noise
(Markov Chain)



$\mathbf{x}_0$ Training Sample

$\mathbf{x}_1$

$\mathbf{x}_2$

$\mathbf{x}_3$

$\mathbf{x}_{T-1}$

$\mathbf{x}_T$ Gaussian Noise (distribution)

Reverse Process (think decoder) [learned]
Learn to recover original image
(i.e. learn to reverse the Markov Chain)

Forward Process

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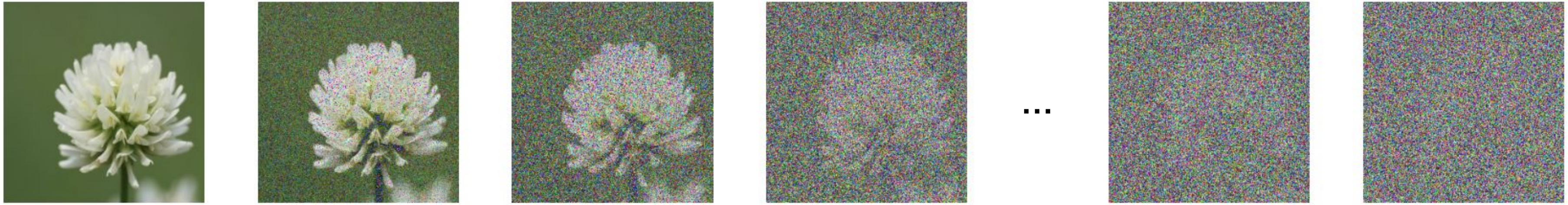
Markov Chain Implications



$\mathbf{x}_0$ $\mathbf{x}_1$ $\mathbf{x}_2$ $\mathbf{x}_3$ ... $\mathbf{x}_{T-1}$ $\mathbf{x}_T$

Direction of dependence

Gaussian Noise: $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$

$q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) = q(\mathbf{x}_t|\mathbf{x}_{t-1})$ T or F?

$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ T or F?

What happens if α□=0 and α□=1?

Details: Forward Process

Can we extend this to sampling $\mathbf{x}_t$ in a closed form?

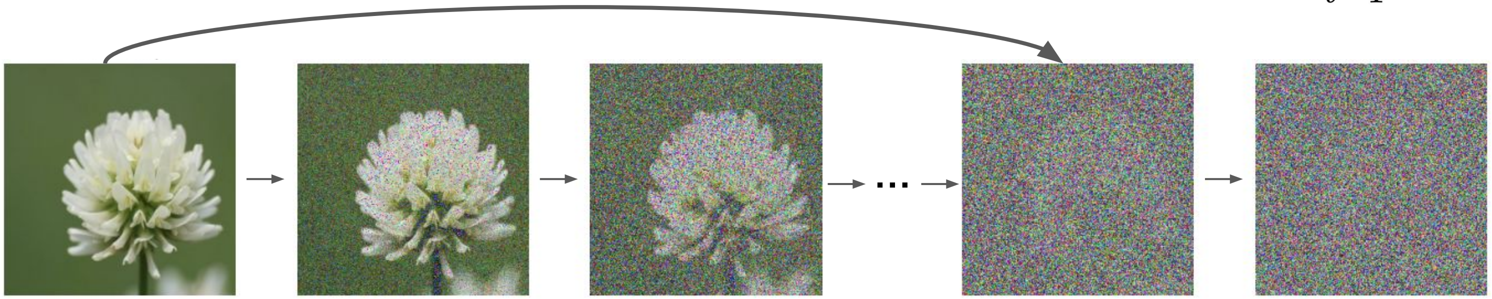
Let $\alpha_t := 1 - \beta_t$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$$

Re-parametrization trick!

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$$

$\boldsymbol{\epsilon}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$



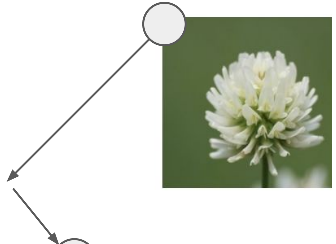
$\mathbf{x}_0$ $\mathbf{x}_1$ $\mathbf{x}_2$ $\mathbf{x}_{T-1}$ $\mathbf{x}_T$

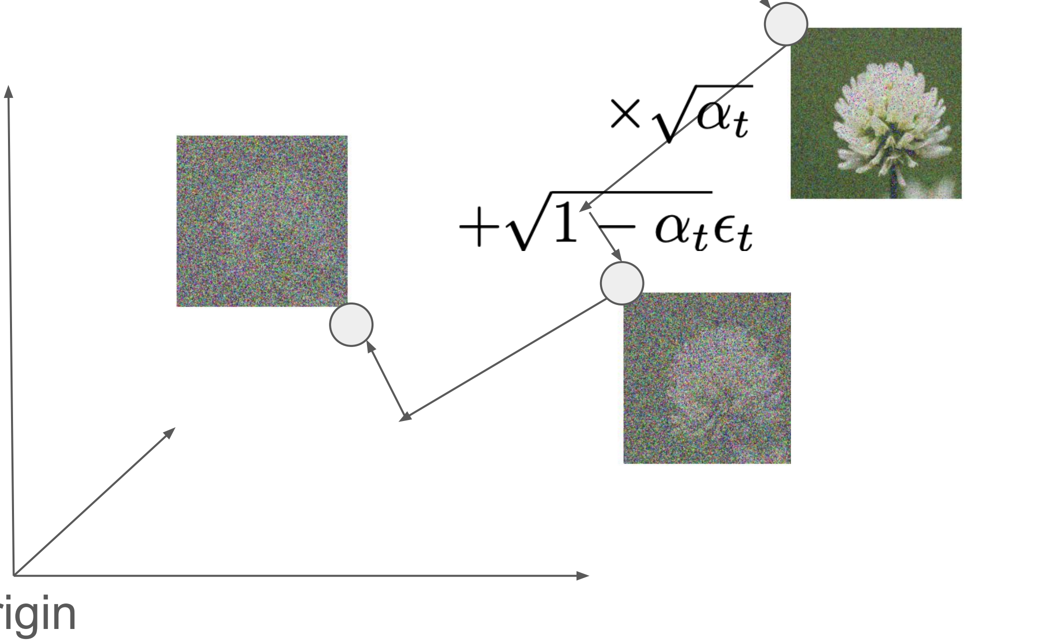
Brownian Motion towards the origin

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$$

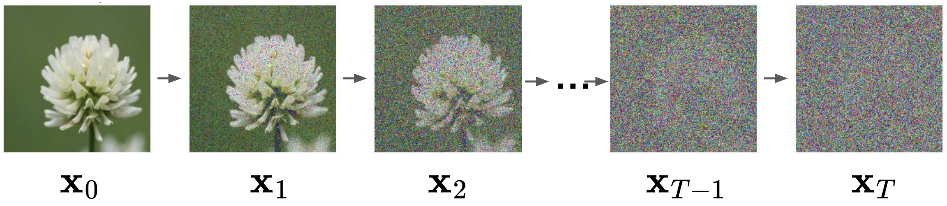
Interpret as two-step move:

1. Scale by $\times\sqrt{\alpha_t}$
2. Add noise $+\sqrt{1-\alpha_t}\epsilon_t$





Details: Forward Process



Inductively, we can say

$$
\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \underline{\sqrt{1-\alpha_t\alpha_{t-1}}\bar{\boldsymbol{\epsilon}}_{t-2}} \\
&= \dots \\
&= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}
\end{aligned}
$$

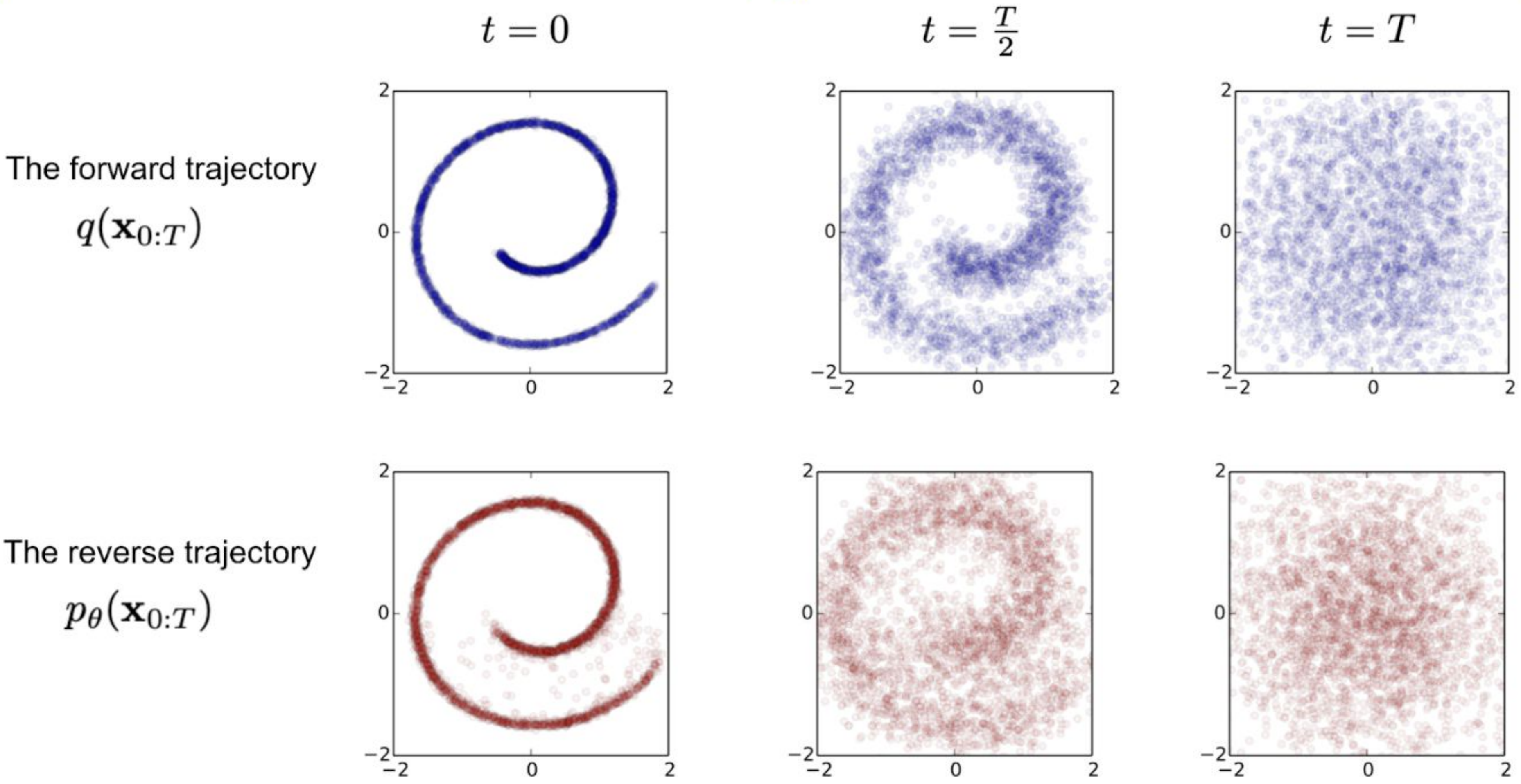
Merged noise.
epsilon is still $\sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

$$\boxed{\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\boxed{\qquad}\mathbf{x}_0, \boxed{\qquad}\mathbf{I})$$

We can sample $q(x_\square \mid x_0)$ in closed form!!!

Reverse Process

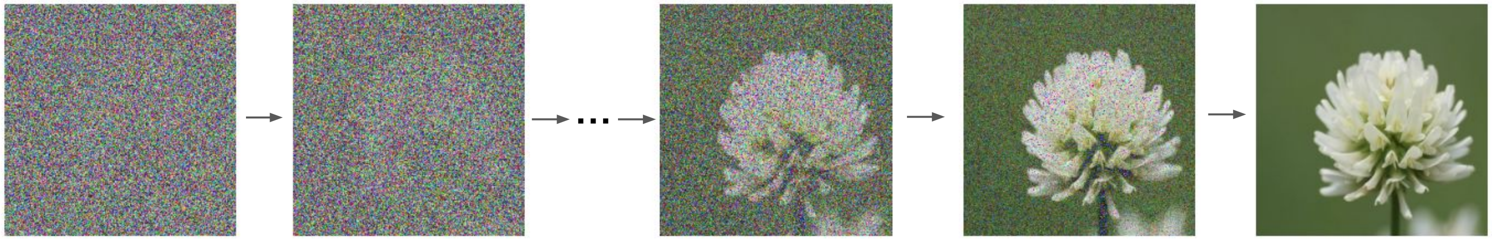


Sohl-Dickstein et al., 2015

Reverse Process Overview

- "Learn to reverse what we just destroyed"
  - Learn time reversal of Markov Chain; we **train a model for this**
- Desired outcome: some $\mathbf{x}_0$ close to the original data distribution
- Problem: $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is **intractable** (can't compute easily)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{\boxed{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\,\boxed{q(\mathbf{x}_{t-1})}}{\boxed{q(\mathbf{x}_t)}}$$



$\mathbf{x}_T$ $\mathbf{x}_{T-1}$ $\mathbf{x}_2$ $\mathbf{x}_1$ $\mathbf{x}_0$

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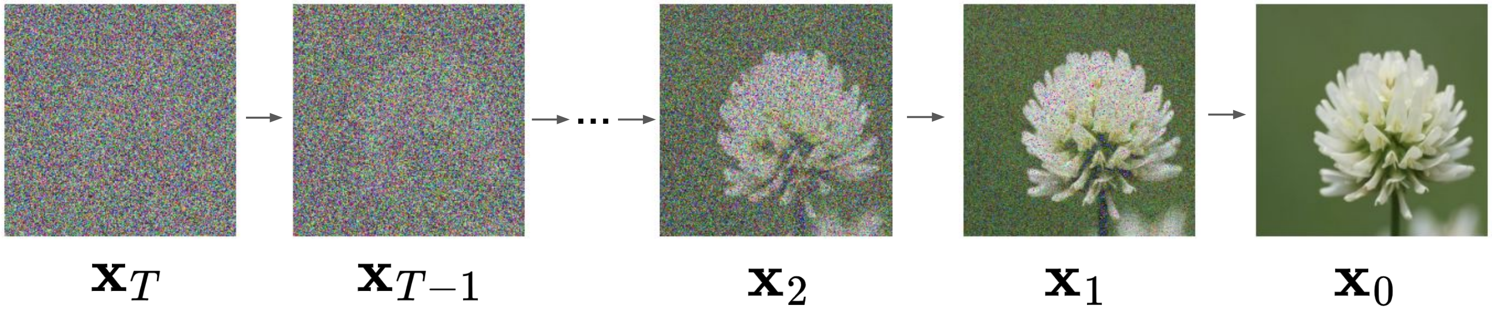
Details: Reverse Process

$q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ Is **not tractable**. Is $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ tractable?

I.e. Can we reverse the forward process given the original data?

$$\boxed{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}}$$



Key Idea

We introduce a generative model to approximate the reverse process $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

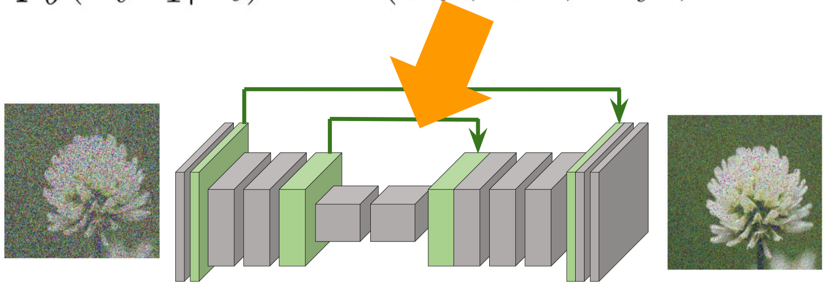
$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$$

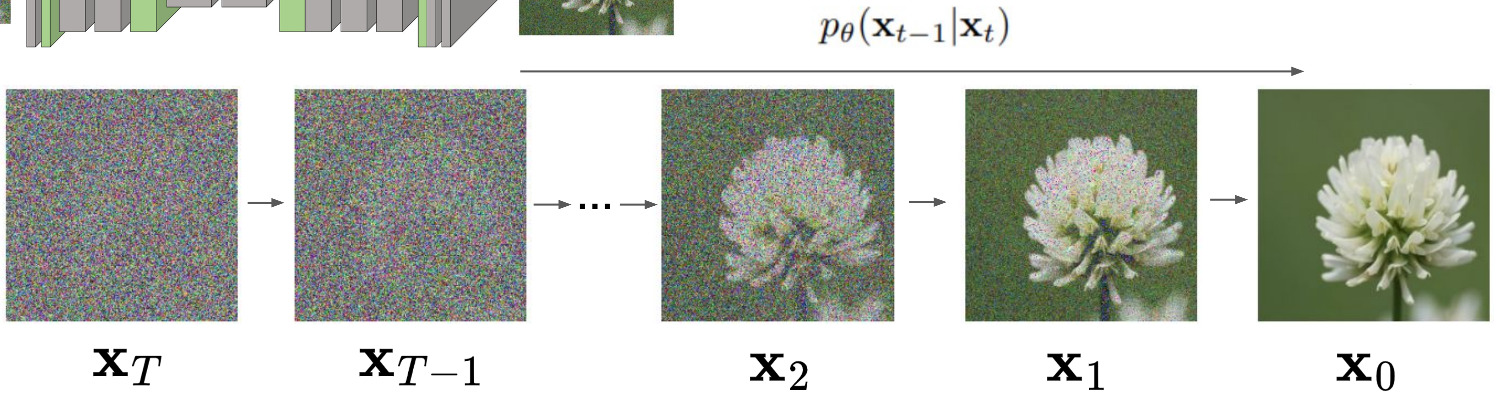
$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I})$$

$$\boxed{\mathbb{E}_{q(\boldsymbol{x}_t|\boldsymbol{x}_0)}\left[D_{\mathrm{KL}}(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \parallel p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t))\right]}$$

Learning Objective!

U-Net





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Reverse process:

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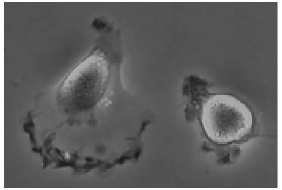
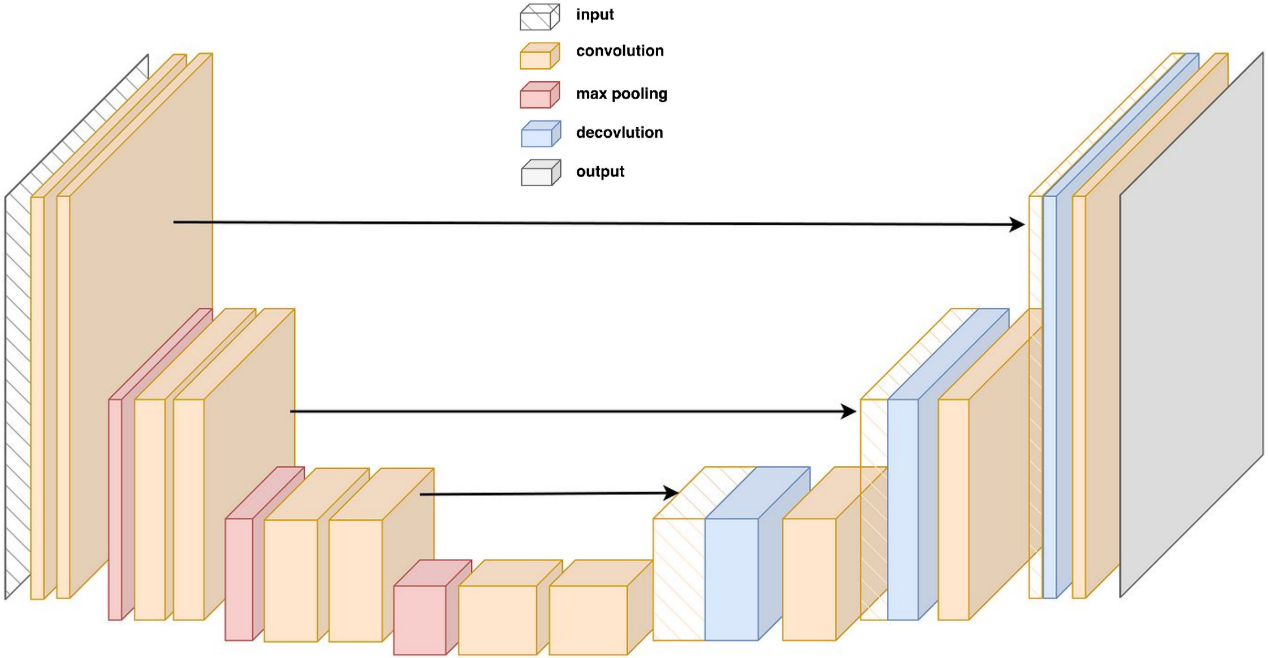
The U-Net

[Ronneberger et al. 2015]





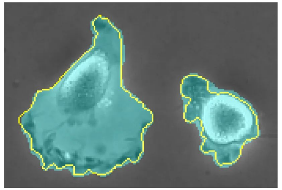


Image Source:
https://miro.medium.com/v2/resize:fit:1400/1*VUS2cCaPB45wcHHFp_fQZQ.png

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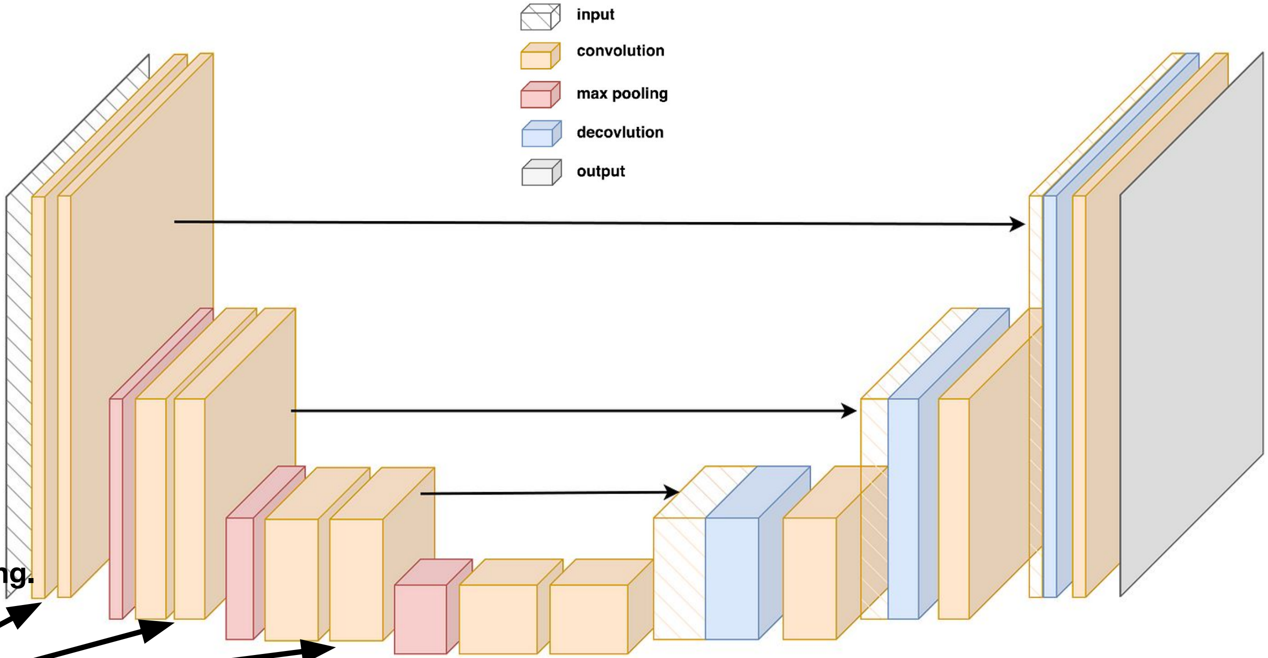
The U-Net

[Ronneberger et al. 2015]

$\sigma = 0.75$

Sinusoidal embedding.

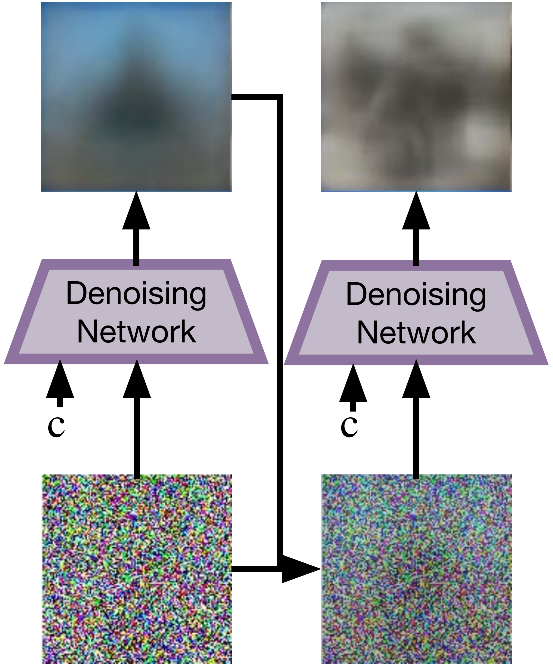


c="An oil painting of a old white man with a white shirt, black hat, and gray side burns."

Image Source:
https://miro.medium.com/v2/resize:fit:1400/1*VUS2cCaPB45wcHHFp_fOZQ.png

c="An astronaut riding a horse"

Diffusion Sampling



$\mathbf{x}_T = \sigma_T \epsilon_T$ $\mathbf{x}_{T-1} = \hat{\mathbf{x}}_\theta + \sigma_{T-1}\epsilon_{T-1}$

$$\boxed{p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \mathbf{x}_\theta(\mathbf{x}_t, t)) \approx q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \mathbf{x}_0)}$$

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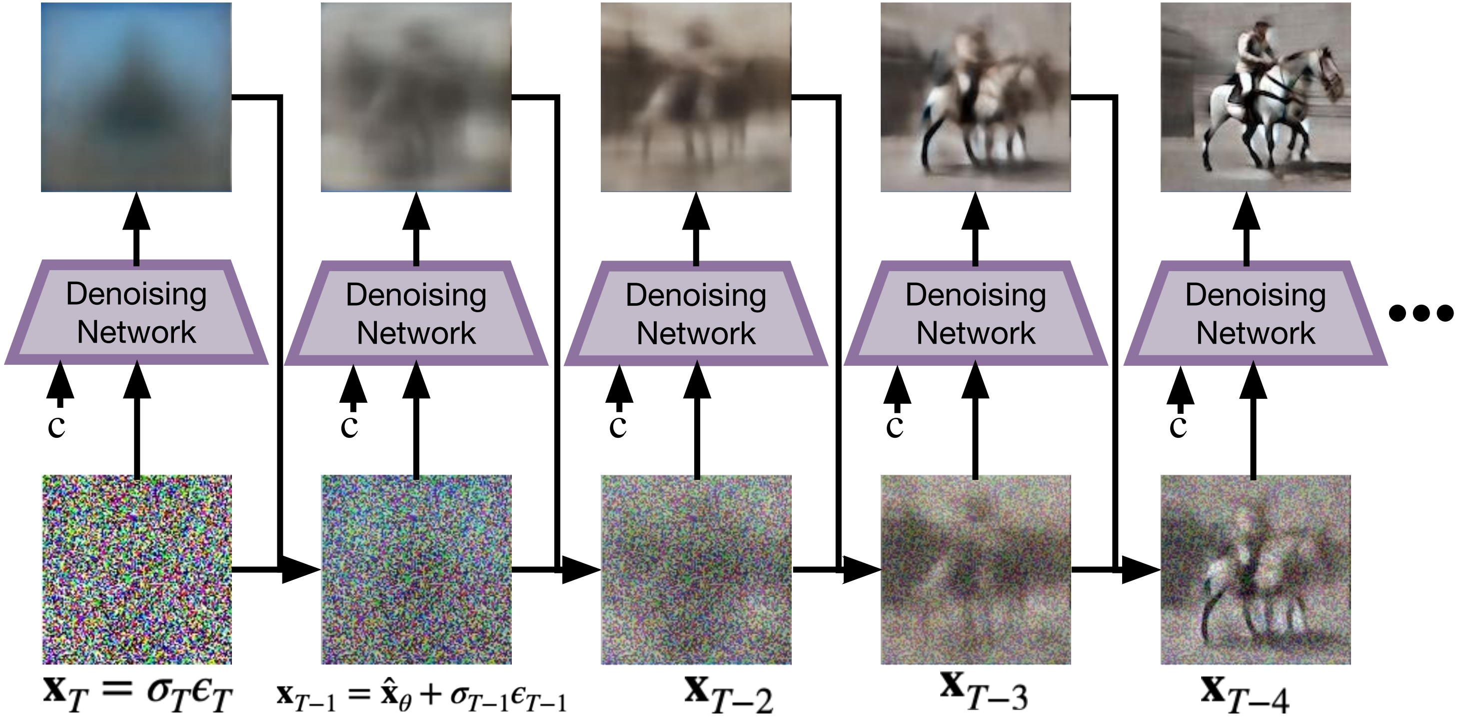
Diffusion Sampling

c="An astronaut riding a horse"



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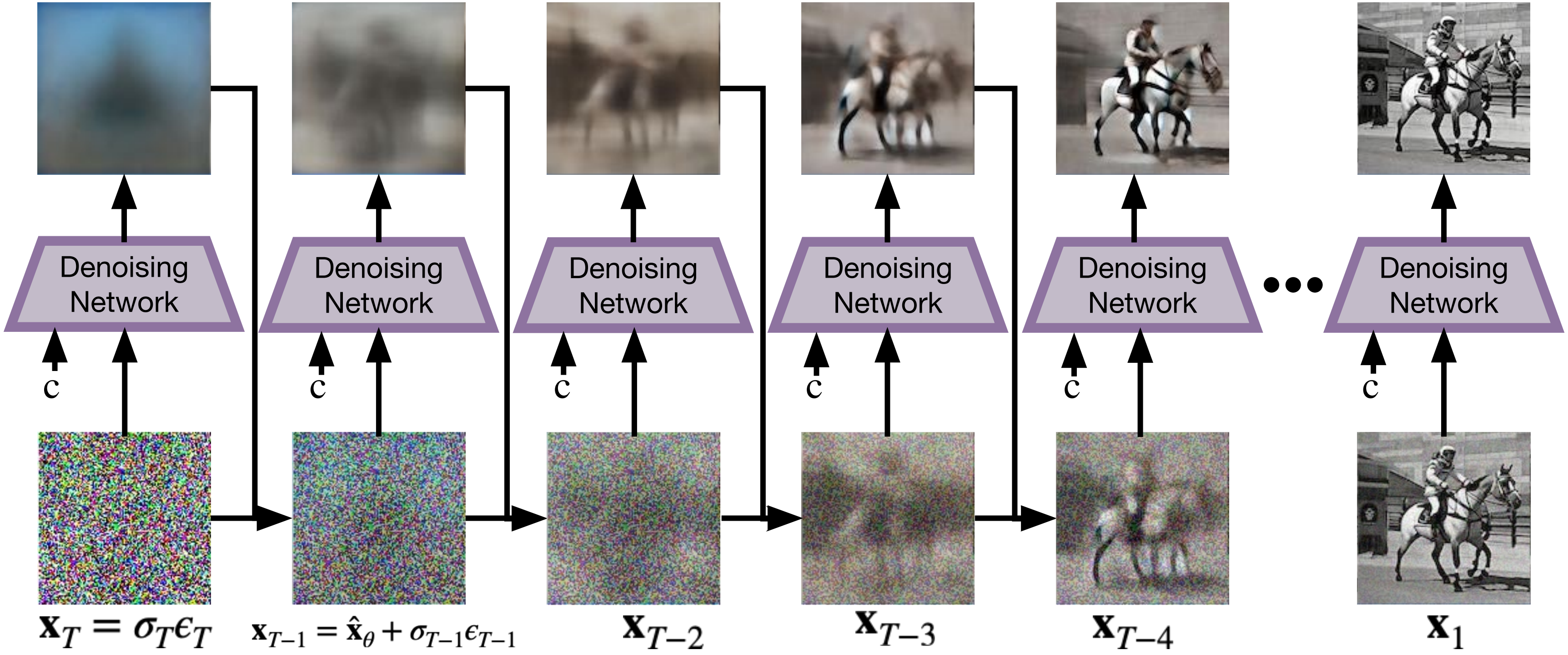
Diffusion Sampling

c="An astronaut riding a horse"



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Diffusion Models

An astronaut riding a horse

Photo Credit: Lin, Shanchuan, et al. "Common diffusion noise schedules and sample steps are flawed." *Proceedings of the IEEE/CVF winter conference on applications of computer vision*. 2024.

Algorithm 1 DDPM Training

repeat
  Sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
  Sample $t \sim \mathcal{U}\{1, 2, \ldots, T\}$
  Sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Compute noisy image:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\,\epsilon$$

  Take optimizer step on:

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2\right]$$

until convergence

Algorithm 1 Deterministic Sampling (DDIM)

Input: trained noise predictor ϵ_θ, noise schedule $\alpha_1, \ldots, \alpha_T$
Sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, T-1, \ldots, 1$ **do**
  Predict noise:

$$\hat{\epsilon} = \epsilon_\theta(\mathbf{x}_t, t)$$

  Estimate $\mathbf{x}_0$:

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\epsilon}}{\sqrt{\bar{\alpha}_t}}$$

  Compute $\mathbf{x}_{t-1}$ deterministically:

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\hat{\epsilon}$$

end for
return $\mathbf{x}_0$

Q: Why is predicting the noise better than predicting the image?

[Jiaming Song, Chenlin Meng, Stefano Ermon ICLR 2021]

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ELBO for Diffusion Models

$$
\begin{aligned}
\log p(\boldsymbol{x}) &\geq \mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\left[\log \frac{p(\boldsymbol{x}_{0:T})}{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\right] \\
&= \mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\left[\log \frac{p(\boldsymbol{x}_T)p_{\boldsymbol{\theta}}(\boldsymbol{x}_0|\boldsymbol{x}_1)}{q(\boldsymbol{x}_1|\boldsymbol{x}_0)} + \log \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)}{\frac{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)\cancel{q(\boldsymbol{x}_t|\boldsymbol{x}_0)}}{\cancel{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0)}}}\right] \\
&= \mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\left[\log \frac{p(\boldsymbol{x}_T)p_{\boldsymbol{\theta}}(\boldsymbol{x}_0|\boldsymbol{x}_1)}{\cancel{q(\boldsymbol{x}_1|\boldsymbol{x}_0)}} + \log \frac{\cancel{q(\boldsymbol{x}_1|\boldsymbol{x}_0)}}{q(\boldsymbol{x}_T|\boldsymbol{x}_0)} + \log \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)}{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)}\right] \\
&= \mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\left[\log \frac{p(\boldsymbol{x}_T)p_{\boldsymbol{\theta}}(\boldsymbol{x}_0|\boldsymbol{x}_1)}{q(\boldsymbol{x}_T|\boldsymbol{x}_0)} + \sum_{t=2}^{T} \log \frac{p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)}{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)}\right] \\
&= \mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}_0|\boldsymbol{x}_1)\right] + \mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\left[\log \frac{p(\boldsymbol{x}_T)}{q(\boldsymbol{x}_T|\boldsymbol{x}_0)}\right] + \sum_{t=2}^{T} \mathbb{E}_{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}\left[\log \frac{p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)}{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)}\right] \\
&= \mathbb{E}_{q(\boldsymbol{x}_1|\boldsymbol{x}_0)}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}_0|\boldsymbol{x}_1)\right] + \mathbb{E}_{q(\boldsymbol{x}_T|\boldsymbol{x}_0)}\left[\log \frac{p(\boldsymbol{x}_T)}{q(\boldsymbol{x}_T|\boldsymbol{x}_0)}\right] + \sum_{t=2}^{T} \mathbb{E}_{q(\boldsymbol{x}_t,\boldsymbol{x}_{t-1}|\boldsymbol{x}_0)}\left[\log \frac{p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)}{q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)}\right] \\
&= \underbrace{\mathbb{E}_{q(\boldsymbol{x}_1|\boldsymbol{x}_0)}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}_0|\boldsymbol{x}_1)\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q(\boldsymbol{x}_T|\boldsymbol{x}_0) \parallel p(\boldsymbol{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\boldsymbol{x}_t|\boldsymbol{x}_0)}\left[D_{\mathrm{KL}}(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0) \parallel p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t))\right]}_{\text{denoising matching term}}
\end{aligned}
$$

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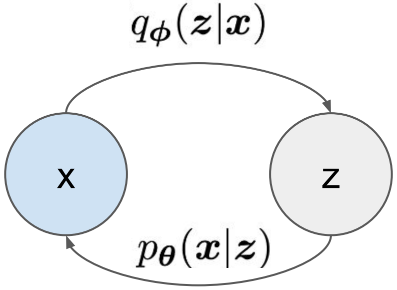
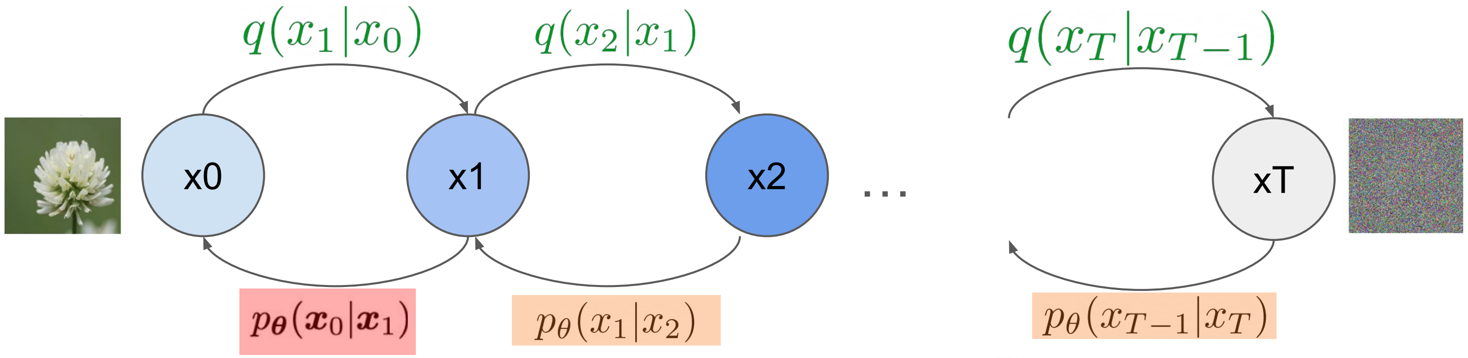
Training Objective

- Bound the likelihood with the ELBO
  - Exactly like VAEs



$$\log p(\boldsymbol{x}) \geq \underbrace{\mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \parallel p(\boldsymbol{z}))}_{\text{prior matching term}}$$



$$\log p(\boldsymbol{x}) \geq \underbrace{\mathbb{E}_{q(\boldsymbol{x}_1|\boldsymbol{x}_0)}\left[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}_0|\boldsymbol{x}_1)\right]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}(q(\boldsymbol{x}_T|\boldsymbol{x}_0) \parallel p(\boldsymbol{x}_T))}_{\text{prior matching term}} - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\boldsymbol{x}_t|\boldsymbol{x}_0)}\left[D_{\mathrm{KL}}(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \parallel p_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t))\right]}_{\text{denoising matching term}}$$

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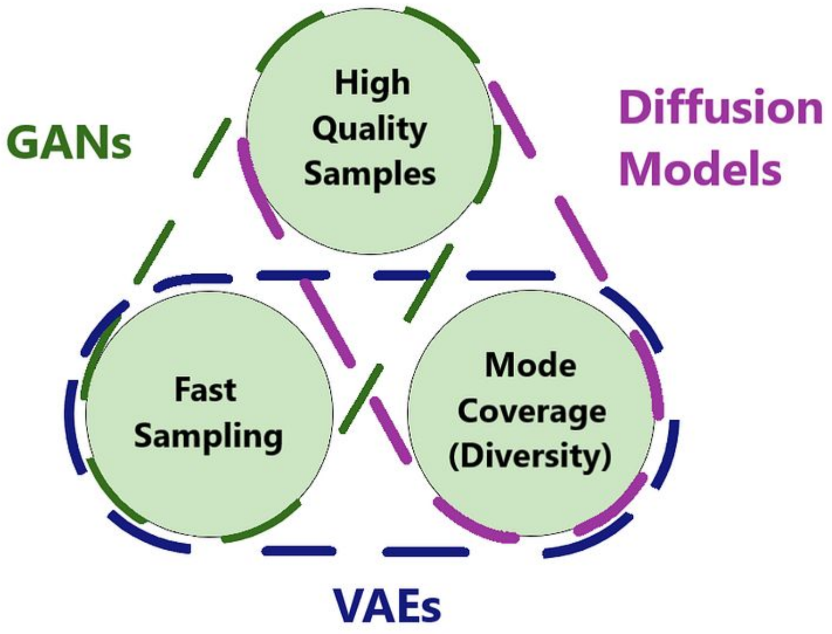
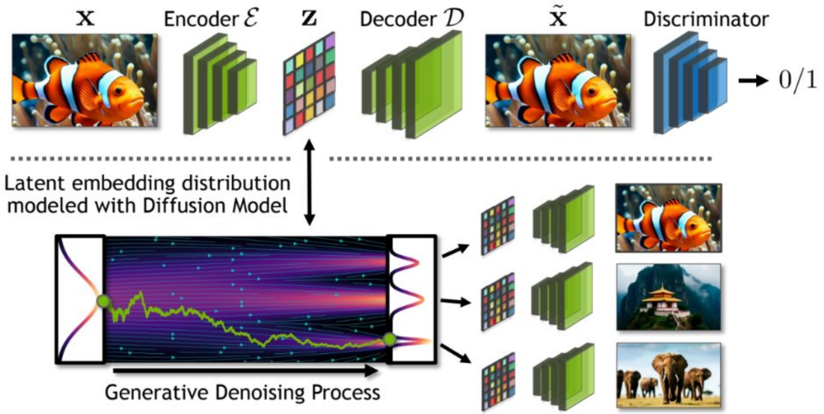
Diffusion Models vs. VAEs vs. GAN

Source



Latent Diffusion

Latent Diffusion Models offer Excellent Trade-off between Performance and Compute Demands



<u>Combines</u>:
Auto-Encoders
Discriminators
Diffusion

➡ LDM with appropriate regularization, compression, downsampling ratio and strong autoencoder reconstruction:

- **Computationally efficient** diffusion model in latent space (compression & lower resolution).
- Yet **very high-performance** (latent diffusion + autoencoder + discriminator = ❤).
- **Highly flexible** (can adjust autoencoder for different tasks and data).

https://neurips2023-ldm-tutorial.github.io/

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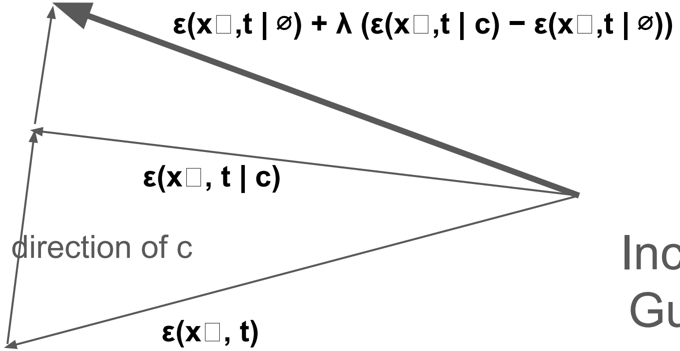
Classifier-Free Guidance

Problem: Diffusion models often ignore the label (i.e. for example the text prompt)

Idea: Give the direction of the conditioning information extra emphasis.
Significantly improves quality of conditional models. Used by practically **every** conditional diffusion model



Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E. L., ... & Norouzi, M. (2022). Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems*, *35*, 36479-36494.

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Video Generation

Sora

Prompt: 3D animation of a small, round, fluffy creature with big, expressive eyes explores a vibrant, enchanted forest.

Prompt: A cat waking up its sleeping owner demanding breakfast.

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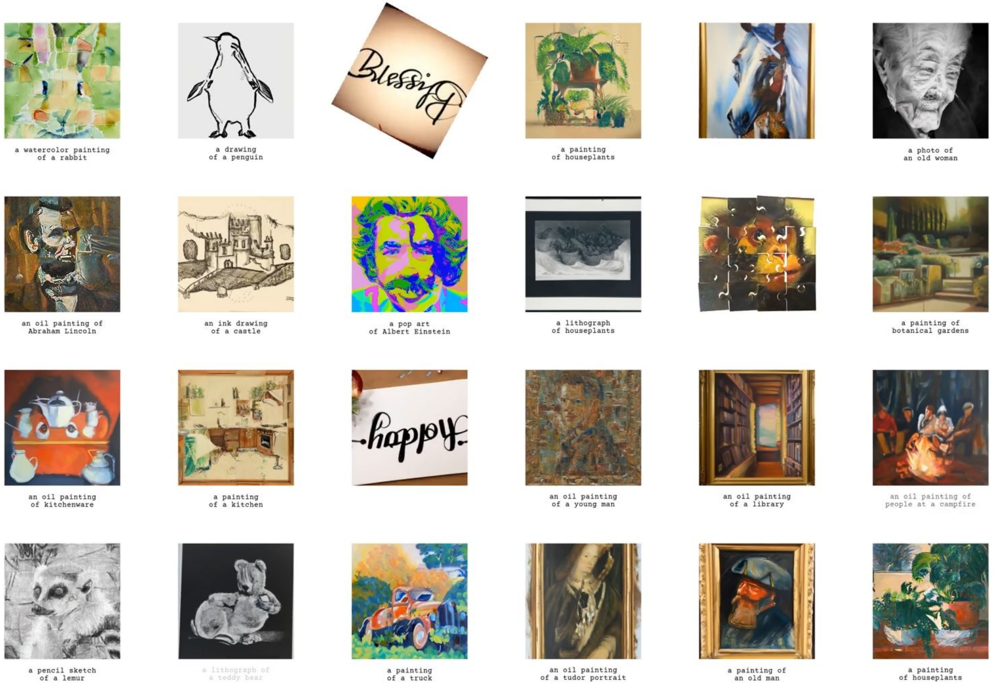
Quiz: How did Prof. Andrew Owens do this?

https://dangeng.github.io/visual_anagrams/



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Overview

- Recap
- Diffusion model overview
- Forward
- Reverse
- Training Objective